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Q1. List top 5 customers by total order amount.

Retrieve the top 5 customers who have spent the most across all sales orders. Show CustomerID, CustomerName, and TotalSpent.

QUERY:

select top 5

c.customerid,

c.name as customername,

sum(so.totalamount) as totalspent

from customer c

join salesorder so

on c.customerid = so.customerid

group by c.customerid, c.name

order by totalspent desc;

	customerid	customername	totalspent
1	98	Armando	232783.99
2	12	Dwight	175719.20
3	84	Courtney	156259.77
4	90	Chelsae	149290.57
5	27	Lawton	141343.95

Q2. Find the number of products supplied by each supplier.

Display SupplierID, SupplierName, and ProductCount. Only include suppliers that have more than 10 products.

select

s.supplierid,

s.name as suppliername,

count(p.productid) as productcount

from supplier s

join purchaseorder p1

on s.supplierid = p1.supplierid

join purchaseorderdetail p2

on p1.orderid = p2.orderid

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join product p
    on p2.productid = p.productid
group by s.supplierid, s.name
having count(p.productid) > 10
order by productcount desc;

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	supplierid	suppliername	productcount
1	7	Kassulke LLC	62
2	8	Lang-Rodriguez	59
3	10	Stokes, Funk and Herman	52
4	2	Bogisich-Block	51
5	3	Abshire Inc	49
6	4	Schmidt LLC	49
7	6	Abernathy-Waelchi	46
8	5	Mann, Will and Fritsch	45
9	1	Schamberger Inc	44
10	9	Sawayn-Adams	43

Q3. Identify products that have been ordered but never returned.

Show ProductID, ProductName, and total order quantity.

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select
    p1.productid,
    p1.name as productname,
    sum(s1.quantity) as totalorderquantity
from product p1
join salesorderdetail s1
    on p1.productid = s1.productid
left join returndetail r1
    on p1.productid = r1.productid
where r1.productid is null
group by p1.productid, p1.name
order by totalorderquantity desc

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	productid	productname	totalorderquantity
1	15	Flecainide Acetate	55
2	3	Sertraline Hydrochloride	20

Q4. For each category, find the most expensive product.

Display CategoryID, CategoryName, ProductName, and Price. Use a subquery to get the max price per category.

select

c1.categoryid, c1.name as categoryname, p.name as productname, p.price

from product p

join category c1

on p.categoryid = c1.categoryid

where p.price = (

select max(p2.price)

from product p2

where p2.categoryid = p.categoryid

)

order by c1.categoryid

	categoryid	categoryname	productname	price
1	2	Landscaping & Irrigation	Betapace AF	1543.64
2	3	Framing (Wood)	ILLUMINATING DAILY MOISTURIZER	2437.99
3	4	Painting & Vinyl Wall Covering	Focalin	2455.81
4	5	Soft Flooring and Base	Isopropyl Alcohol Surgical Scrub	2554.94
5	6	Electrical and Fire Alarm	Fluoxetine	726.23
6	7	Fire Protection	Oral Analgesic	2980.75
7	9	Termite Control	Secura Moisturizing Cleanser	2627.59
8	10	Exterior Signage	Flecainide Acetate	2325.87
9	12	Asphalt Paving	Acetaminophen PM	2677.44
10	13	NULL	COMETRIQ	2778.36
11	14	Glass & Glazing	Losartan Potassium	2749.20
12	15	Hard Tile & Stone	Quetiapine Fumarate	1831.84
13	16	Elevator	Sertraline Hydrochloride	2896.17
14	17	Termite Control	SohMed Cold Relief	2813.71
15	18	Wall Protection	Natural Cherry Honey Herb Throat Drops	2502.47
16	19	Epoxy Flooring	Entex T	2012.38
17	20	Hard Tile & Stone	Rutalital, Anatinophen and Caffeine	85.95

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Q5. List all sales orders with customer name, product name, category, and supplier.

For each sales order, display:

OrderID, CustomerName, ProductName, CategoryName, SupplierName, and Quantity.

select

s.orderid, c.name as customername, p.name as productname, c2.name as categoryname,

s2.name as suppliername,

s3.quantity

from salesorder s

join customer c

on s.customerid = c.customerid

join salesorderdetail s3

on s.orderid = s3.orderid

join product p

on s3.productid = p.productid

join category c2

on p.categoryid = c2.categoryid

join purchaseorderdetail p2

on p.productid = p2.productid

join purchaseorder p3

on p2.orderid = p3.orderid

join supplier s2

on p3.supplierid = s2.supplierid

order by s.orderid

	orderid	customername	productname	categoryname	suppliername	quantity
1	1	Armando	Focalin	Painting & Vinyl Wall Covering	Abernathy-Waelchi	12
2	1	Armando	Focalin	Painting & Vinyl Wall Covering	Stokes, Funk and Herman	12
3	1	Armando	Focalin	Painting & Vinyl Wall Covering	Abshire Inc	12
4	1	Armando	Focalin	Painting & Vinyl Wall Covering	Sawayn-Adams	12
5	1	Armando	Focalin	Painting & Vinyl Wall Covering	Bogisich-Block	12
6	1	Armando	Focalin	Painting & Vinyl Wall Covering	Kassulke LLC	12
7	1	Armando	Focalin	Painting & Vinyl Wall Covering	Stokes, Funk and Herman	12
8	1	Armando	Focalin	Painting & Vinyl Wall Covering	Kassulke LLC	12
9	1	Armando	Focalin	Painting & Vinyl Wall Covering	Schmidt LLC	12
10	1	Armando	Focalin	Painting & Vinyl Wall Covering	Sawayn-Adams	12
11	1	Armando	Focalin	Painting & Vinyl Wall Covering	Abshire Inc	12
12	1	Armando	Focalin	Painting & Vinyl Wall Covering	Abshire Inc	12
13	1	Armando	Focalin	Painting & Vinyl Wall Covering	Bogisich-Block	12
14	1	Armando	Focalin	Painting & Vinyl Wall Covering	Sawayn-Adams	12
15	1	Armando	Focalin	Painting & Vinyl Wall Covering	Abernathy-Waelchi	12
16	1	Armando	Focalin	Painting & Vinyl Wall Covering	Abshire Inc	12
17	2	Armando	Acadlmetaine, NILLI	Painting & Vinyl Wall Covering	Lana-Bodineus	10

Query executed successfully. DESKTOP-009EPOL\SQLEXPRESS ... DESKTOP-009EPOL\Dell (56) NEXT_COLA_OLTP 00:00:00 Row: 1, Col: 1 1,265 rows

Q6. Find all shipments with details of warehouse, manager, and products shipped.

Display:

ShipmentID, WarehouseName, ManagerName, ProductName, QuantityShipped, and TrackingNumber.

select

s.shipmentid,

l.name as warehousename,

e.name as managername,

p.name as productname,

s2.quantity as quantityshipped,

s.trackingnumber

from shipment s

join warehouse w

on s.warehouseid = w.warehouseid

join employee e

on w.managerid = e.employeeid

join location l

on w.locationid = l.locationid

join shipmentdetail s2

on s.shipmentid = s2.shipmentid

join product p

on s2.productid = p.productid

order by s.shipmentid

	shipmentid	warehousename	managename	productname	quantityshipped	trackingnumber
1	1	Ms	Daloris	Midol Complete	23	NULL
2	2	Rev	Rose	Quetiapine Fumarate	25	NULL
3	3	Ms	Daloris	Secura Moisturizing Cleanser	23	NULL
4	4	Rev	Rose	Entex T	14	NULL
5	5	Ms	Daloris	Promethazine Hydrochloride	29	NULL
6	6	Mr	Neille	PASPALLUM NOTATUM POLLEN	6	NULL
7	7	Mr	Aurora	SHISEIDO ADVANCED HYDRO-LIQUID COMPACT (REFILL)	1	NULL
8	8	Mr	Neille	Icy Hot Back and Large Areas	8	NULL
9	9	Rev	Rose	SHISEIDO ADVANCED HYDRO-LIQUID COMPACT (REFILL)	7	NULL
10	10	Rev	Rose	COMETRIQ	14	NULL
11	11	Mr	Aurora	Natural Cherry Honey Herb Throat Drops	20	NULL
12	12	Rev	Rose	Icy Hot Back and Large Areas	30	NULL
13	13	Ms	Daloris	ILLUMINATING DAILY MOISTURIZER	18	NULL
14	14	Ms	Daloris	Midol Complete	18	NULL
15	15	Mr	Aurora	Maximum Strength	15	NULL
16	16	Ms	Daloris	Oral Analgesic	17	NULL
17	17	Mr	Rose	Entex T	22	NULL

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Q7. Find the top 3 highest-value orders per customer using RANK(). Display CustomerID, CustomerName, OrderID, and TotalAmount.

select *

from (

select

c.customerid, c.name as customername, s.orderid, s.totalamount,

rank() over (

partition by c.customerid

order by s.totalamount desc

) as orderrank

from customer c

join salesorder s

on c.customerid = s.customerid

) ranked_orders

where orderrank <= 3

order by customerid, orderrank;

	shipmentid	warehousename	managername	productname	quantityshipped	trackingnumber
1	1	Ms	Daloris	Midol Complete	23	NULL
2	2	Rev	Rose	Quetiapine Fumarate	25	NULL
3	3	Ms	Daloris	Secura Moisturizing Cleanser	23	NULL
4	4	Rev	Rose	Entex T	14	NULL
5	5	Ms	Daloris	Promethazine Hydrochloride	29	NULL
6	6	Mr	Neille	PASPALUM NOTATUM POLLEN	6	NULL
7	7	Mr	Aurora	SHISEIDO ADVANCED HYDRO-LIQUID COMPACT (REFILL)	1	NULL
8	8	Mr	Neille	Icy Hot Back and Large Areas	8	NULL
9	9	Rev	Rose	SHISEIDO ADVANCED HYDRO-LIQUID COMPACT (REFILL)	7	NULL
10	10	Rev	Rose	COMETRIQ	14	NULL
11	11	Mr	Aurora	Natural Cherry Honey Herb Throat Drops	20	NULL
12	12	Rev	Rose	Icy Hot Back and Large Areas	30	NULL
13	13	Ms	Daloris	ILLUMINATING DAILY MOISTURIZER	18	NULL
14	14	Ms	Daloris	Midol Complete	18	NULL
15	15	Mr	Aurora	Maximum Strength	15	NULL
16	16	Ms	Daloris	Oral Analgesic	17	NULL
17	17	Mr	Rose	Entex T	22	NULL

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Q8. For each product, show its sales history with the previous and next sales quantities (based on order date). Display ProductID, ProductName, OrderID, OrderDate, Quantity, PrevQuantity, and NextQuantity.

select

p.productid, p.name as productname, s.orderid, s.orderdate, sd.quantity,

lag(sd.quantity) over (

partition by p.productid

order by s.orderdate

) as prevquantity,

lead(sd.quantity) over (

partition by p.productid

order by s.orderdate

) as nextquantity

from salesorderdetail sd

join salesorder s

on sd.orderid = s.orderid

join product p

on sd.productid = p.productid

order by p.productid, s.orderdate;

	productid	productname	orderid	orderdate	quantity	prevquantity	nextquantity
1	1	Butalbital, Acetaminophen and Caffeine	79	2003-04-21 04:11:35	19	NULL	17
2	1	Butalbital, Acetaminophen and Caffeine	61	2016-05-18 03:37:40	17	15	13
3	1	Butalbital, Acetaminophen and Caffeine	88	2017-07-11 14:02:27	13	17	NULL
4	2	SHISEIDO ADVANCED HYDRO-LIQUID COMPACT (REFILL)	30	2000-12-26 15:50:29	23	NULL	NULL
5	3	Sertaline Hydrochloride	76	2009-10-24 02:52:28	20	NULL	NULL
6	4	SohMed Cold Relief	72	2003-11-14 11:53:28	24	NULL	14
7	4	SohMed Cold Relief	44	2004-04-04 09:26:01	14	24	11
8	4	SohMed Cold Relief	55	2012-04-24 07:46:10	11	14	34
9	4	SohMed Cold Relief	70	2021-08-30 09:18:44	34	11	NULL
10	5	Promethazine Hydrochloride	15	2000-05-29 09:37:55	13	NULL	23
11	5	Promethazine Hydrochloride	80	2017-08-05 16:48:19	23	13	NULL
12	6	Ondansetron	23	2002-02-06 08:48:16	14	NULL	24
13	6	Ondansetron	86	2011-05-18 11:07:42	24	14	37
14	6	Ondansetron	53	2013-01-29 02:06:34	37	24	35
15	6	Ondansetron	50	2020-10-08 05:33:26	35	37	NULL
16	7	TopCare Antacid Calcium	43	2007-12-19 07:08:49	15	NULL	29
17	7	TopCare Antacid Calcium	04	2008-03-01 03:58:21	29	15	NULL

Query executed successfully. DESKTOP-009EPOL\SQLXPRESS ... DESKTOP-009EPOL\Dell (56) NEXT_COLA_OLTP 00:00:00 Row: 1, Col: 1 100 rows

Q9. Create a view named vw_CustomerOrderSummary that shows for each customer:

CustomerID, CustomerName, TotalOrders, TotalAmountSpent, and LastOrderDate.

create view view_Customer_Order_Summary as

select

c.CustomerID,

c.Name AS CustomerName,

count(so.OrderID) as TotalOrders,

coalesce(SUM(so.TotalAmount), 0) as TotalAmountSpent,

max(so.OrderDate) as LastOrderDate

from customer c

left join salesorder so

on c.CustomerID = so.CustomerID

group by c.CustomerID, c.Name

Commands completed successfully.

Completion time: 2026-06-07T22:03:22.3022397+05:00

select * from view_Customer_Order_Summary

	CustomerID	CustomerName	TotalOrders	TotalAmountSpent	LastOrderDate
1	1	Pavla	0	0.00	NULL
2	2	Daune	1	29549.27	2008-12-11 11:36:26
3	3	Neville	2	99452.77	2010-11-30 13:33:08
4	4	Eustace	1	29705.50	2019-12-22 19:57:18
5	5	Elga	0	0.00	NULL
6	6	Murray	1	12849.15	2008-07-06 01:10:56
7	7	Lem	1	73192.12	2012-03-18 16:38:10
8	8	Reube	0	0.00	NULL
9	9	Sergeant	0	0.00	NULL
10	10	Valeria	0	0.00	NULL
11	11	Dory	1	51043.81	2007-12-19 07:08:49
12	12	Dwight	3	175719.20	2020-07-21 05:01:11
13	13	Liliane	1	44724.06	2012-12-21 02:35:15
14	14	Juditha	0	0.00	NULL
15	15	Claudius	0	0.00	NULL
16	16	Sayres	1	22896.23	2022-11-16 00:14:29
17	17	Joselle	0	0.00	NULL

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Q10. Write a stored procedure sp_GetSupplierSales that takes a SupplierID as input and returns the total sales amount for all products supplied by that supplier.

SELECT

s.SupplierID,

s.Name AS SupplierName,

SUM(sod.Quantity * p.Price) AS TotalSales

FROM Supplier s

JOIN PurchaseOrder p

ON s.SupplierID = p.SupplierID

JOIN PurchaseOrderDetail p2

ON p.OrderID = p2.OrderID

JOIN Product p

ON p2.ProductID = p.ProductID

JOIN SalesOrderDetail s2

ON p.ProductID = s2.ProductID

 Messages

Msg 208, Level 16, State 1, Line 187
Invalid object name 'Supplier'.

Completion time: 2026-06-07T22:13:00.9338145+05:00